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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ADAPTIVE SOFTWARE ENGINEERING PROJECT (24CI3201)
A.Y. 2026 - 2027

TITLE OF THE PROJECT	**Adaptive Academic Project Collaboration Portal**	
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Abstract

Academic institutions often encounter significant challenges in managing student research and academic projects due to fragmented communication, manual record keeping, and the lack of a centralized collaboration portal. Students face difficulties in finding project teammates, submitting proposals, tracking milestones, and monitoring project progress. Faculty guides are required to supervise multiple projects simultaneously, making it difficult to review submissions and provide timely feedback. Project coordinators also experience challenges in allocating faculty guides, monitoring project status, scheduling reviews, and generating reports efficiently. Existing project collaboration systems primarily focus on basic project tracking and lack adaptive workflow capabilities that can automatically respond to changes during the project lifecycle.

To address these challenges, this project proposes an **Adaptive Academic Project Collaboration Portal (AAPCP)**, a web-based application that streamlines the complete academic project lifecycle through predefined rule-based adaptive workflows. Unlike conventional systems, the proposed portal automatically assigns faculty guides based on workload, prevents duplicate project topics, updates project status when milestones are completed, sends deadline reminders, escalates delayed projects to the coordinator, and automatically generates progress reports. These adaptive features improve collaboration, reduce manual intervention, and ensure efficient project execution without the use of Artificial Intelligence.

The proposed portal is developed using **React.js** for the frontend, **Spring Boot (Java)** for the backend, and **MySQL** as the database, while **JWT (JSON Web Token)** provides secure authentication and role-based access control. The development process follows the **Agile Scrum** methodology, enabling iterative development through user stories, sprint planning, backlog management, and continuous stakeholder feedback. Modern DevOps practices are incorporated using **Git, GitHub, Docker, Kubernetes (Minikube), and CI/CD pipelines** to support automated building, testing, deployment, and scalable application management. Application security is enhanced through secure coding practices, **OWASP** security guidelines, and role-based authorization.

The portal supports three major stakeholders: **Students, Faculty Guides, and Project Coordinators**. Students can register, submit project proposals, create or join teams, upload project documents, manage tasks, and monitor milestones. Faculty guides can review proposals, assign milestones, evaluate submissions, provide feedback, and track team progress. Project coordinators can manage project batches, allocate faculty members, monitor ongoing projects, schedule reviews, generate reports, and administer user accounts. The adaptive workflow continuously responds to project events and predefined business rules, enabling efficient collaboration and timely completion of academic projects.

The proposed system is evaluated based on functional correctness, usability, response time, security, reliability, adaptive workflow efficiency, and user satisfaction. The expected outcomes include improved collaboration among students and faculty, balanced faculty workload, reduced project delays, enhanced project monitoring, and a scalable portal that demonstrates the practical application of **Software Engineering, Agile Development, DevOps, DevSecOps, and Cloud Computing** concepts in academic project collaboration.

Signature of the Guide HOD